PS 3: Week 9

From Samples to Populations

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Agenda

1. Final Projects
2. Generalizability and Weighting
3. Descriptive Hypothesis Testing

Housekeeping

- Congrats on finishing the midterm!
- Data collection or surveys are due **tonight**: *October 23rd, 11:59 pm*

Final Projects – Choosing a Project

- Which proposal is the most feasible?
- What's the most interesting to the most members?
- General notes to consider once you've chosen a project:
  - Make sure you have a "Prefer not to answer" or similar option for sensitive questions
    - Examples of sensitive topics can include, but are not limited to sexual orientation and GPA
  - Consider adding additional variables for HTEs or descriptive hypothesis tests

Final Projects – Data Collection

- **Non-survey-based**
  - Upload as a CSV or Excel file
  - Extensions allowed, but make sure you have a plan to collect data
- **Survey-based**
  - Instructions on how to make a Google Form are on the "Data Collection" assignment on bCourses, including how to randomize treatment/control
  - I will have comments to your groups on the surveys by October 26th, please make the changes by **Thursday, October 29th**
  - I will have my sections complete the surveys from the other section on October 30th and November 2nd so that you have the data with plenty of time to submit the first draft of the report by **November 13th**
- You're welcome to work on this during office hours

Attendance Survey



From Samples to Populations

- **Experimental**
  - We run an experiment exposing people to campaign materials and measure the effect of these materials on someone's likelihood of voting for our candidate
    - How do we know this generalizes across cases?
- **Descriptive**
  - We run a poll measuring support for our candidate
    - How do we know this generalizes across our population?

Weighting

Category	Weight	Student's Grade	Weighted Grade
In-class Individual Assignments	0.10	75%	7.50%
In-class Group Assignments	0.05	90%	4.50%
Problem Sets	0.30	95%	28.50%
Final Project	0.10	90%	9.00%
Peer Evaluations	0.05	100%	5.00%
Section Attendance	0.05	100%	5.00%
Midterm	0.15	90%	13.50%
Final	0.20	90%	18.00%
Total			**91.00%**

Weighted grade = weight $\times$ grade, summed across categories.

Generalizing From Populations

- We take samples from a population because we want to learn something about that population
- But samples are always prone to some kind of sampling bias!
  - **Sample selection bias:** who do we ask to take the survey?
  - **Non-response bias:** what kind of people take (or don't take) political surveys?
- And what about random chance?
  - E.g. a sample I take on a Monday will be different from a sample I take on a Tuesday
- You already have the tools to address these problems, what are they?

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- You already have the tools to address these problems, what are they?
  - **Sampling bias:** weighting, probability sampling
  - **Random chance:** 95% confidence intervals, increasing sample size, standard error, p-values

Descriptive Hypothesis Testing: Is this Random Chance?

- We collected some data and found that Group A is different from Group B.
  - Ex.: Younger voters support our candidate more than older voters.
- There is still the same issue as before:
  - Is the observed difference between Group A and Group B due to random chance (from sampling), or is it because the two groups truly are not the same?
- Note: Our goal with this approach is to describe differences, **NOT** determine causality! We would need to do an experiment to prove that (remember Heterogeneous Treatment Effects)

Descriptive Hypothesis Testing w/ Weights: Addressing Sampling Bias

- Some people are more likely to answer our surveys/partake in our experiments than others
  - We can weight groups that are underrepresented in our sample *more* to provide better balance in our data
  - We can weight groups that are overrepresented in our sample *less* to provide better balance in our data
  - Ex: if college-educated voters are overrepresented in our sample, they get a smaller weight; if non-college-educated voters are underrepresented in our sample, they get a bigger weight
- Adding weights to our descriptive hypothesis tests can help us take into account sampling bias when comparing two groups

Heterogeneous Treatment Effects

- "Heterogeneous" = different
  - vs the word "homogeneous" = same
- The general idea:
  - A treatment could have different effects on different groups in terms of magnitude (size of effect)

Group	Love for Oski before seeing photo	Love for Oski after seeing photo	Difference (after - before)
Berkeley students	75%	100%	+25%
Stanford students	70%	80%	+10%

Heterogeneous Treatment Effects: Interpretation and Limitations

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  - We don't randomly assign Berkeley vs. Stanford
  - Remember: Causal statements pertain to what we can randomize
  - We can randomize the treatment, but not the characteristics that we choose to base our subsets on

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  - We don't randomly assign Berkeley vs. Stanford
  - Remember: Causal statements pertain to what we can randomize
  - We can randomize the treatment, but not the characteristics that we choose to base our subsets on
- HTEs just tell us how the treatment effect varies within groups – they don't suggest that group characteristics cause anything in our outcome

Acknowledgements

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